

Exact diff Eqn

$Mdx + Ndy = 0$ is called exact if $\exists f(x,y)$ st.

$$d(f(x,y)) = Mdx + Ndy.$$

* $Mdx + Ndy = 0$ is exact differential Eqn iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

* solution of Exact diff Eqn

$$\int M dx + \int \left(\begin{array}{c} \text{terms in } N, \text{ Not} \\ \text{containing } x \end{array} \right) dy = C$$

y constant

Rule I :- Guess I.F.

Rule II :- $Mdx + Ndy = 0$ is not Exact
 $\hookrightarrow$ Homogeneous Equation
 $Mx + Ny \neq 0$, then $\frac{1}{Mx + Ny}$ is I.F.

Rule III :- Let $Mdx + Ndy = 0$ is not Exact diff eqn
 and Eqn is of the form

$$f_1(xy) y dx + f_2(xy) x dy = 0$$

Then I.F. $\frac{1}{Mx - Ny}$ if $Mx - Ny \neq 0$ function of 'xy'

Example :- $y(1+xy)dx + x(1-xy)dy = 0$

Solution :-

$$M = y + xy^2$$

$$N = x - x^2y$$

$$\frac{\partial M}{\partial y} = 1 + 2xy$$

$$\frac{\partial N}{\partial x} = 1 - 2xy$$

Not Exact.

$$\frac{\partial}{\partial y} - \dots = 0$$

Not Exact.

dx

$$Mx - Ny = xy + x^2y^2 - xy + x^2y^2 = 2x^2y^2 \neq 0$$

$$\text{I.F.} \quad \frac{1}{2x^2y^2}$$

Exact diff Equation :-

$$\left(\frac{1}{2x^2y} + \frac{1}{2x} \right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y} \right) dy = 0$$

$$M = \frac{1}{2x^2y} + \frac{1}{2x}$$

$$N = \frac{1}{2xy^2} - \frac{1}{2y}$$

$$\frac{\partial M}{\partial y} = \frac{-1}{2x^2y^2} \quad \text{exact}$$

$$\frac{\partial N}{\partial x} = \frac{-1}{2x^2y^2}$$

Solution is

$$\int \left(\frac{1}{2x^2y} + \frac{1}{2x} \right) dx + \int -\frac{1}{2y} dy = C$$

y constant

$$\frac{1}{2y} \cdot \frac{-1}{x} + \frac{1}{2} \ln|x| + \left(-\frac{1}{2} \ln|xy| \right) = C$$

$$\frac{-1}{2xy} + \frac{1}{2} \ln \left| \frac{x}{y} \right| = C.$$

Rule IV & V :-

$Mdx + Ndy = 0$ is not an exact diff

(iv)

If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x alone, say $f(x)$

then $e^{\int f(x) dx}$ is I.F of $Mdx + Ndy = 0$

(V) If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y alone, say $f(y)$

then $e^{\int f(y) dy}$ is I.f. of $Mdx + Ndy = 0$

Example:- $(x^2 + y^2 + 2x)dx + 2ydy = 0$

$$M = x^2 + y^2 + 2x \quad N = 2y$$

$$\frac{\partial M}{\partial y} = 2y \quad \frac{\partial N}{\partial x} = 0$$

Not Exact

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{2y}{2y} = 1$$

I.f. :- $e^{\int 1 dx} = e^x$

$$e^x (x^2 + y^2 + 2x)dx + e^x \cdot 2y dy = 0$$

$$M = e^x (x^2 + y^2 + 2x) \quad N = 2e^x y$$

$$\frac{\partial M}{\partial y} = e^x (2y) \quad \frac{\partial N}{\partial x} = 2e^x y$$

exact.

Solution

$$\int (e^x x^2 + e^x y^2 + 2xe^x) dx = C$$

y constant

$$\boxed{\int e^x x^2 dx} + e^x y^2 + \int 2xe^x dx = C$$

$$x^2 e^x - \int e^x \cdot 2x dx + e^x y^2 + \int 2xe^x dx = C$$

$$x^2 e^x + e^x y^2 = C$$

Example:-

$$(2xy^4 e^y + 2xy^3 + y) dx + (x^2 y^4 e^y - x^2 y^2 - 3x) dy = 0$$

$$M = 2xy^4 e^y + 2xy^3 + y$$

$$N = x^2 y^4 e^y - x^2 y^2 - 3x$$

$$\frac{\partial M}{\partial y} = 2xy^4 e^y + 8xy^3 e^y + 6xy^2 + 1$$

$$\frac{\partial N}{\partial x} = 2xy^4 e^y - 2xy^2 - 3$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -8xy^3 e^y - 8xy^2 - 4$$

$$= -4(2xy^3 e^y + 2xy^2 + 1)$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\frac{4}{y} M$$

$$\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = -\frac{4}{y} \text{ is function of } y$$

$$\text{I.I.} \quad e^{\int -\frac{4}{y} dy} = e^{-4 \ln |y|} = \frac{1}{y^4}$$

$$* \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3} \right) dx + \left(x^2 e^y - \frac{x^2}{y^2} - \frac{3x}{y^4} \right) dy = 0$$

$$M = 2xe^y + \frac{2x}{y} + \frac{1}{y^3}$$

$$N = x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}$$

$$\frac{\partial M}{\partial y} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}$$

exact

$$\frac{\partial N}{\partial x} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}$$

Solution

$$\int \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3} \right) dx = C \Rightarrow x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = C$$

y constant

Rule VI :- If $Mdx + Ndy = 0$ is not exact diff Eqn and it is of the form

$$x^\alpha y^\beta (my dx + nx dy) = 0$$

OR

$$x^\alpha y^\beta (my dx + nx dy) + x^{\alpha'} y^{\beta'} (m'y dx + n'x dy) = 0$$

where $\alpha, \beta, \alpha', \beta', m, n, m', n'$ all are constants.

then I.F. is of the type $x^h y^k$, where we choose h, k such that that $x^h y^k (Mdx + Ndy) = 0$ is exact diff Eqn.

Example:-

$$(4xy^2 + 6y) dx + (5x^2y + 8x) dy = 0$$

$$M = 4xy^2 + 6y$$

$$N = 5x^2y + 8x$$

$$M = 4xy^2 + 6y$$

$$\frac{\partial M}{\partial y} = 8xy + 6$$

$$N = 5x^2y + 8x$$

$$\frac{\partial N}{\partial x} = 10xy + 8$$

Not Exact

$$4xy^2 dx + 6y dx + 5x^2y dy + 8x dy = 0$$

$$xy(4y dx + 5x dy) + 1(6y dx + 8x dy) = 0$$

$$\text{I.F} = x^h y^k$$

Equation x I.F.

$$x^h y^k (4xy^2 + 6y) dx + x^h y^k (5x^2y + 8x) dy = 0$$

$$M = x^h y^k (4xy^2 + 6y)$$

$$M = 4x^{h+1} y^{k+2} + 6x^h y^{k+1}$$

$$N = x^h y^k (5x^2y + 8x)$$

$$N = 5x^{h+2} y^{k+1} + 8x^{h+1} y^k$$

$$\frac{\partial M}{\partial y} = 4(k+2) \underline{x^{h+1} y^{k+1}} + 6(k+1) \underline{x^h y^k}$$

$$\frac{\partial N}{\partial x} = 5(h+2) \underline{x^{h+1} y^{k+1}} + 8(h+1) \underline{x^h y^k}$$

Eqn will be exact

$$\text{if } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\Rightarrow \left. \begin{aligned} 4(k+2) &= 5(h+2) \\ 6(k+1) &= 8(h+1) \end{aligned} \right\}$$

$$\boxed{h=2, k=3}$$

$$4k - 5h = 2$$

$$6k - 8h = 2$$

I.f. $x^2 y^3$

Exact diff eqn

$$x^2 y^3 (4x y^2 + 6y) dx + (x^2 y^3) (5x^2 y + 8x) dy = 0$$

$$(4x^3 y^5 + 6x^2 y^4) dx + (5x^4 y^4 + 8x^3 y^3) dy = 0$$

$$M = 4x^3 y^5 + 6x^2 y^4$$

$$N = 5x^4 y^4 + 8x^3 y^3$$

$$\frac{\partial M}{\partial y} = 20x^3 y^4 + 24x^2 y^3$$

$$\frac{\partial N}{\partial x} = 20x^3 y^4 + 24x^2 y^3$$

Exact

Solution

$$\int (4x^3 y^5 + 6x^2 y^4) dx = C$$

$y \text{ constant}$

$$x^4 y^5 + 2x^3 y^4 = C$$

* Show that $x^4 y^5 + 2x^3 y^4 = C$ is a

general solution of

$$(4xy^2 + 6y) dx + (5x^2 y + 8x) dy = 0$$

* $\underbrace{d(x^4 y^5 + 2x^3 y^4)}_f = 0$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$= (4x^3y^5 + 6x^2y^4)dx + (5x^4y^4 + 8x^3y^3)dy = 0$$

$$x^2y^3 (4xy^2 + 6y)dx + (5x^2y + 8x)dy = 0$$